CPE 486/586 Classwork 05: Gradient Descent, Max/Min, Expression Graph

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 23, 2026 (In-Class)

1. Find the minimals and maximal value of $f(x) = 3x - x^3$ on the interval $[-1, 3]$. **(6 Points)**

Answer

2. Find out the minimized for a function $f(x) = (x-3)^2$ using gradient descent $x_{k+1} = x_k - \eta \nabla f(x_k)$ for three iterations and fill the table below. Choose learning rate $\eta = 0.1$. Start at $x_0 = 0$. After three iteration how far is x from its true solution? See next page for the table. **(6+1 Points)**

Answer

k	x_k	$f'(x_k)$	$x_{k+1} = x_k - 0.1\, f'(x_k)$	$f(x_k)$
0	0	-6	0.6	9
1				
2				
3				

3. Draw an expression graph for $C = (4x - 7)^2 + yx$ and use chain rule to evaluate $\frac{\partial C}{\partial x}$. I only need expression of the chain rule because I am not providing you actual values of x and y. Note that x appears multiple times, so there would be a plus in final chain rule. **(7 Points)**

Answer